

Agam: An AI-Driven Cognitive Distortion Detection System for Continuous Mental Health Monitoring and Therapy Support

Jai Krishna Prasath¹, Alagu Manikandan S²

^{1,2}Department of Computer Science and Engineering (Artificial Intelligence & Data Science)

^{1,2}Sri Venkateswara College of Engineering, Chennai, Tamil Nadu.

¹2023ad0676@svce.ac.in, ²2023ad0007@svce.ac.in

ABSTRACT

since mental health therapy is intermittent, recurrence and patient attrition are common problems in depression treatment. Patients are nevertheless susceptible to emotional decline in between sessions due to rumination, negative thought patterns, and maladaptive interpretations because treatment sessions are usually only held once a week. Early diagnosis of cognitive impairment and ongoing monitoring are hampered by this gap. Agam, an AI-powered mental wellness system for ongoing cognitive monitoring and assistance, is presented in this study. To identify cognitive distortions in real time, Agam combines machine learning, natural language processing, and the concepts of cognitive behavioral therapy (CBT). A broad language model facilitates structured cognitive reframing through guided engagement, while a refined transformer-based BERT model examines user diary inputs. In order to detect relapse risks, the system also monitors behavioral signs and gives clinicians useful information. Agam facilitates proactive mental health intervention in between sessions and enhances therapeutic continuity.

KEYWORDS

Artificial Intelligence in Mental Health, Cognitive Behavioral Therapy, Cognitive Distortion Detection, Natural Language Processing, Digital Therapeutics, Continuous Monitoring, Therapist Decision Support

I. INTRODUCTION

An increasing number of people worldwide are concerned about mental health issues like depression, anxiety, and stress-related disorders, which have a substantial negative influence on people's productivity, quality of life, and general well-being. Even with advances in psychotherapy, especially Cognitive Behavioral Therapy (CBT), the discontinuous nature of service delivery remains a key barrier. While emotional distress and maladaptive cognitive habits change constantly in daily life, therapy sessions are usually held at predetermined intervals, usually once a week. People often have habitual negative thoughts, rumination, and cognitive distortions including overgeneralization, emotional reasoning, and catastrophizing in between therapy sessions. Mental health disorders are largely maintained and made worse by these cognitive misconceptions. However, retrospective self-reports given during sessions are frequently the only way for physicians to gain episodic insight into a patient's mental state. This gap delays prompt action and hinders the early diagnosis of declining cognitive patterns.

Numerous systemic issues, such as inadequate symptom relief, high relapse rates, patient attrition, and loss of follow-up, are made worse by this discontinuity. According to research, a sizable percentage of patients stop attending therapy early or on a regular basis, frequently as a result of the lack of ongoing support and feedback systems. Systems that allow for the real-time monitoring of behavioral and cognitive data and extend therapeutic care beyond clinical encounters are therefore desperately needed. The study of human language to identify psychological patterns and emotional states has been made possible by recent developments in Artificial Intelligence (AI), especially in Natural Language Processing (NLP) and transformer-based models like BERT. Furthermore, chatbots with AI capabilities have shown promise in providing scalable mental health assistance. However, current systems lack formal mechanisms for recognizing and correcting cognitive distortions, which are essential to cognitive behavioral therapy (CBT), and instead mostly rely on generic conversational approaches or mood tracking. This study suggests Agam, an AI-powered mental wellness system that offers structured psychological support and ongoing cognitive monitoring in between therapy sessions, as a solution to these drawbacks. Agam presents an innovative classifier-first architecture that uses user-generated journaling data to identify cognitive distortions using a refined BERT model. Through a procedure known as the "Cognitive Courtroom," a Large Language Model (LLM) processes these identified cognitive signals to enable structured therapeutic encounters. Instead of using generic conversational outputs, this method guarantees that responses are based on CBT concepts. Agam also combines predictive modeling and behavioral analytics to find early signs of therapeutic disengagement and relapse. Data-driven therapeutic decision-making is made possible by the system's therapist-facing dashboard, which offers aggregated insights into patient cognitive trends. The following are this work's main contributions:

- i. Transformer-based NLP models are used in a novel framework for detecting cognitive distortion.
- ii. Cognitive Courtroom is an organized AI-driven intervention system that adheres to CBT concepts.
- iii. an ongoing monitoring system that combines behavioral and cognitive cues to identify risks early.
- iv. a layer of clinical intelligence to provide therapists with useful information in between sessions.

Agam seeks to close the gap between episodic treatment and ongoing mental health care by converting subjective cognitive experiences into quantifiable signals, allowing for proactive, individualized, and scalable psychological support.

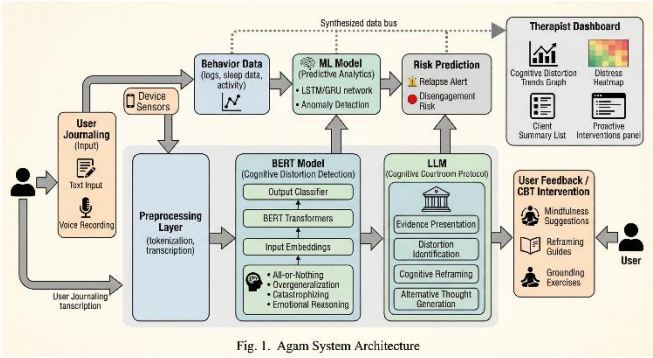


Figure 1 : Agam System Architecture

II. RELATED WORK

In recent years, there has been a lot of interest in the use of artificial intelligence (AI) to mental health treatment, especially in the creation of digital therapy tools and conversational agents. Early research in this field concentrated on chatbots driven by artificial intelligence (AI) that used conversational interfaces to convey aspects of cognitive behavioral therapy (CBT). The viability of offering scaled mental health support has been proven by systems like Woebot and Wysa, with studies showing a decrease in users' symptoms of anxiety and sadness [1]–[5]. Although these systems provide constant accessibility and user involvement, their capacity to thoroughly examine cognitive patterns is constrained by their primary reliance on predefined dialog flows or rule-based frameworks.

The application of large language models (LLMs) and generative AI to mental health applications has been investigated recently. The user experience and engagement are enhanced by these platforms, which provide more organic and context-aware dialogues [2]. Despite these advancements, a lot of LLM-based systems continue to produce answers without a formal foundation in psychological frameworks, which raises questions regarding therapeutic validity, consistency, and dependability. Specifically, they frequently lack the tools necessary to specifically recognize and correct cognitive distortions, which are essential to cognitive behavioral therapy.

The use of Natural Language Processing (NLP) tools to identify mental health signs from textual data has been the subject of parallel study. Research has demonstrated the ability to deduce emotional states, stress levels, and depression indications from language patterns in user-generated content, such as journal entries or social media posts [12], [15]. Because of their capacity for contextual awareness, transformer-based models—in particular, BERT—have shown excellent performance in evaluating such textual material. These methods make it possible to automatically identify psychological signs, but they are usually restricted to categorization tasks and do not offer structured therapy interventions.

The identification of cognitive distortions by machine learning methods has been the subject of several studies. For example, transformer-based designs have been used in recent

research to categorize distortions from textual inputs, including overgeneralization, emotional reasoning, and catastrophizing [12]. Although these methods confirm that it is possible to computationally identify dysfunctional thought patterns, they frequently function as independent systems without being included into a more comprehensive therapeutic framework. Systematic studies of AI-based chatbots for mental health also point out important shortcomings in current systems, such as limited integration with healthcare providers, lack of customisation, and inadequate clinical foundation [9]–[11]. The majority of systems either concentrate on symptom tracking or user engagement, but they fall short of offering a complete system that integrates behavioral monitoring, organized intervention, cognitive analysis, and physician support.

Additionally, studies on digital mental health platforms highlight the significance of early intervention and ongoing monitoring. Activity levels, sleep patterns, and engagement patterns are examples of behavioral signals that have been found to be significant markers of mental health state. Few systems, nonetheless, successfully combine these signals with cognitive-level analysis to allow for proactive response.

In conclusion, even though research on AI-based mental health assistance has advanced significantly, there is still a key gap in the development of systems that integrate continuous behavioral monitoring, structured CBT-based intervention, cognitive distortion detection, and therapist collaboration. In order to close this gap, this study presents Agam, a revolutionary system that enables ongoing and proactive mental health care by combining transformer-based cognitive analysis with organized therapeutic contact and clinical intelligence.

Feature	Chatbots	NLP Models	Agam
CBT Support	✓	✗	✓
Cognitive Distortion Detect	✗	✓	✓
Structured Intervention	✗	✗	✓
Continuous Monitoring	✓	✗	✓
Therapist Integration	✗	✗	✓

Table 1 : Comparison of Existing Systems and Proposed Approach

III. PROPOSED SYSTEM

This study offers Agam, an AI-driven platform that enables continuous cognitive monitoring and structured

psychological intervention in between therapy sessions, as a solution to the shortcomings of current mental health systems. Large Language Models (LLMs), transformer-based Natural Language Processing (NLP), behavioral analytics, and clinician feedback systems are all integrated into the system's hybrid architecture. Agam takes a classifier-first, intervention-second approach, making sure that all interactions are based on clinically relevant cognitive patterns, in contrast to traditional chatbot systems that rely on generic conversational responses. The general architecture of the system is intended to provide therapists with practical insights while simultaneously capturing, analyzing, and reacting to user cognitive and behavioral signals in real time.

A. System Overview

Agam is made up of four main parts:

1. User Interaction Layer
2. Cognitive Analysis Layer
3. Intervention Layer (Cognitive Courtroom)
4. Clinical Intelligence and Monitoring Layer

Together, these elements convert unstructured user input into tailored interventions and structured cognitive insights.

B. User Interaction Layer

The main interface between the user and the system is the user interaction layer. Through text or voice journaling, users interact with Agam, allowing them to communicate ideas, feelings, and experiences in natural language.

This strategy guarantees:

- Capturing automatic ideas in real time
- input with lots of context that reflects everyday cognitive tendencies
- decreased reliance on past memories during treatment sessions

Speech-to-text processing ensures consistent downstream analysis by converting voice inputs to text.

C. Cognitive Analysis Layer

Finding cognitive distortions in user input is the responsibility of the cognitive analysis layer. This is accomplished by deploying a refined transformer-based model (BERT) on-device to guarantee data security, low latency, and privacy.

Key Functions:

- Text preprocessing and tokenization
- Contextual embedding generation using BERT
- Multi-class classification of cognitive distortions

The model detects distortions such as:

- All-or-nothing thinking
- Overgeneralization
- Emotional reasoning
- Catastrophizing
- Labeling
- "Should" statements
- Personalization

This layer serves as the basis for downstream reasoning and intervention by transforming unstructured thoughts into structured cognitive signals.

D. Intervention Layer: Cognitive Courtroom

The "Cognitive Courtroom," a structured dialogue mechanism influenced by CBT techniques, is a novel concept introduced in the intervention layer.

Workflow:

- The user shares a troubling idea.
- The related cognitive distortion is recognized by the system.
- The LLM has a guided conversation about reflection.

During this procedure:

- The AI takes on the function of a logical assessor (evidence prosecutor).
- The user's idea is regarded as a claim that is being investigated.
- The system uses CBT methods like:
 - Examining the evidence
 - Restructuring cognition
 - Re-attribution
 - Cost-benefit evaluation
 - Considering gray areas

Unlike traditional chatbots, the responses are:

- Structured
- Clinically grounded
- Context-aware

This guarantees that the system promotes active cognitive reframing and metacognitive awareness rather than offering general confidence.

E. Behavioral Monitoring and Risk Prediction

Agam uses behavioral analytics in addition to cognitive cues to identify early indicators of disengagement or relapse.

Input Signals (with user consent):

- Journaling frequency
- Sleep patterns
- Physical activity levels
- App engagement trends

Machine learning models are used to process these signals in order to determine:

- Declining engagement
- Increased negative language intensity
- Behavioral withdrawal patterns

Output:

- Risk scores for **relapse** or **attrition**
- Early warning triggers for intervention

F. Clinical Intelligence Layer

A therapist-facing dashboard that combines insights from behavioral and cognitive data is provided by the Clinical Intelligence and Monitoring Layer. In between therapy sessions, this layer is intended to assist physicians in keeping track of patients' progress and making well-informed decisions.

The dashboard allows therapists to track changes in thought patterns over time by displaying longitudinal trends in cognitive distortions. Additionally, it records temporal behavioral patterns that might provide important context for clinical evaluation, such as times of increased distress (such late-night emotional spikes). By identifying patients who need urgent care based on calculated risk levels, the

technology also makes patient prioritization easier. The platform creates automatic session summaries and facilitates integration with therapist-defined frameworks, enabling customisation depending on particular therapeutic techniques, to further support clinical processes. This layer serves as a clinical decision support system by combining these skills. It allows therapists to watch patients continually outside of planned sessions, spot subclinical warning signs early, and provide more prompt and individualized therapies.

G. System Workflow Summary

The complete system operates as follows:

User inputs journaling data (text/voice)

1. BERT model detects cognitive distortions
2. LLM generates structured CBT-based responses
3. Behavioral data is analyzed for risk prediction
4. Insights are shared with both:
 - User (intervention)
 - Therapist (dashboard)

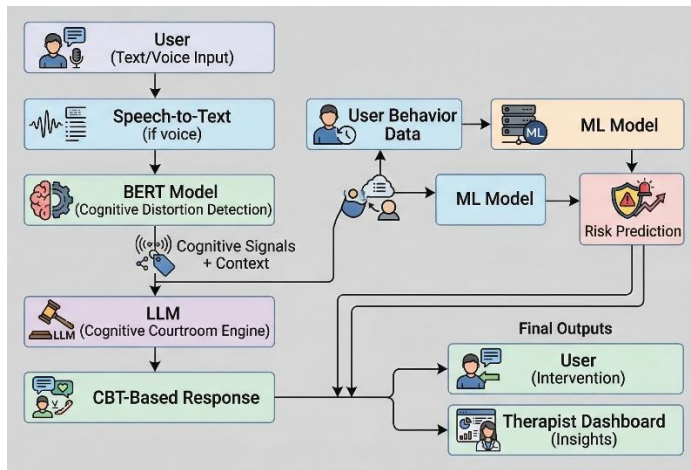


Figure 2 : Proposed Agam System Architecture

IV. METHODOLOGY

The technological technique for the suggested Agam system, which includes behavioral risk prediction, structured intervention creation, and cognitive distortion detection, is presented in this part. The system uses a multi-stage pipeline that combines machine learning methods, huge language models, and transformer-based natural language processing.

A. Data Collection and Preprocessing

Agam's main input source is user-generated journaling data, which includes both text-based entries and voice inputs that are translated into text using speech-to-text conversion. This makes it possible to continuously and naturally record user thoughts in an unrestricted format.

While preparing the data for model input, the preprocessing pipeline is made to preserve contextual meaning. The following are the main preprocessing steps:

- Text normalization (handling punctuation and lowercasing)

- Tokenization with the BERT tokenizer
- Elimination of noise and unnecessary symbols
- Maintaining context for semantic comprehension

This approach preserves contextual dependencies, which are essential for detecting cognitive distortions, in contrast to conventional sentiment analysis techniques.

B. Cognitive Distortion Detection using BERT

A refined Bidirectional Encoder Representations from Transformers (BERT) model for multi-class classification of cognitive aberrations is the basis of Agam. A fully connected classification layer with softmax activation comes after a pre-trained BERT base model in the architecture.

Several embedding representations are used to encode each journaling entry, such as:

- Embeddings of tokens
- Segment embeddings
- Embeddings in position

All-or-nothing thinking, overgeneralization, mental filtering, emotional reasoning, catastrophizing, labeling, personalization, and "should" statements are among the cognitive distortion categories into which the model divides inputs.

The training objective is defined using cross-entropy loss:

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

where y_i denotes the true label and $\hat{y}_i$ represents the predicted probability. The model outputs both the predicted distortion category and a confidence score, enabling structured interpretation of user cognition.

C. Structured Cognitive Intervention using LLM

The system uses a Large Language Model (LLM) to produce structured treatment responses based on Cognitive Behavioral Therapy (CBT) principles after distortion identification. The original journaling input, contextual user history, identified distortion kind, and optional therapist-defined frameworks are all sent to the LLM. The Cognitive Courtroom framework, which treats the user's idea as a claim under examination, is used to implement the intervention mechanism. The system leads the user through a methodical process of reasoning that consists of:

- Evidence-based questioning
- Identification of cognitive distortions
- Generation of alternative interpretations
- Guided cognitive reframing

A claim like "I always fail at everything," for instance, might be categorized as overgeneralization. The approach then promotes cognitive remodeling by encouraging the identification of counterexamples and reflective questioning. This guarantees that answers stay structured, context-aware, and clinically based.

D. Behavioral Signal Processing and Risk Prediction

Agam uses behavioral analytics in addition to cognitive analysis to find early signs of disengagement or relapse. Journaling frequency, sentiment trends, sleep habits (if available), and application engagement metrics are just a few

of the behavioral cues that the system takes into account. A time-series feature vector is used to depict these signals:

$$X = [f_1, f_2, f_3, \dots, f_n]$$

Where a behavioral indication is associated with each feature. Depending on the implementation, the predictive models that are employed could be logistic regression, random forest, or long short-term memory (LSTM) networks.

The following outputs are generated by the system:

- Probability of relapse
- Probability of disengagement

An intervention is triggered when the predicted risk exceeds a predefined threshold:

$$P(\text{risk}) > \theta$$

Proactive detection of at-risk users and prompt assistance are made possible by this method.

E. System Integration Pipeline

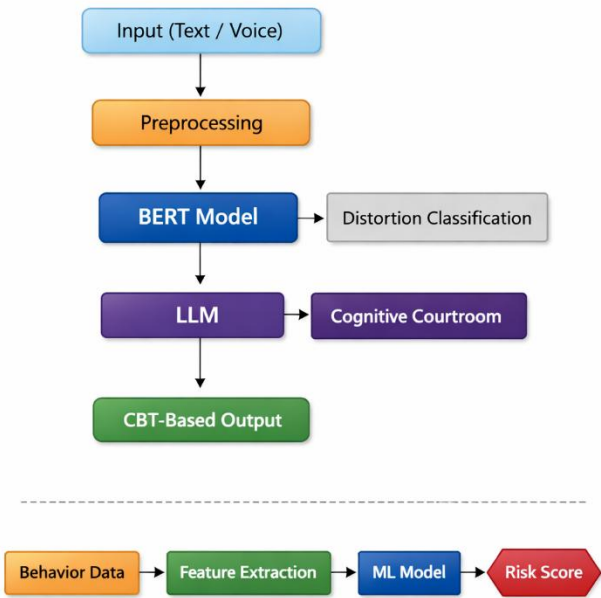


Figure 3 : Methodology Pipeline of Agam

The entire Agam system functions as a multi-stage pipeline that combines elements of behavioral and cognitive processing. The following steps make up the workflow:

- Input acquisition through text or voice journaling
- Data preprocessing
- Cognitive distortion detection using BERT
- Structured intervention generation using LLM
- Behavioral signal analysis and risk prediction
- Output generation and feedback delivery

Scalability, real-time processing capability, and smooth system component interaction are all guaranteed by this modular pipeline.

Cognitive Distortion Classes

Distortion Type	Description
All-or-Nothing	Black-and-white thinking
Overgeneralization	Generalizing from one event
Emotional Reasoning	Believing feelings as facts
Labeling	Assigning negative identity
Catastrophizing	Expecting worst outcomes
Should Statements	Rigid expectations

Table 2 : Cognitive Distortion Classes

V. RESULTS AND EVALUATION

The evaluation strategy and anticipated results of the suggested Agam system are presented in this section. Three main modules are evaluated because of the multi-component architecture: behavioral risk prediction, intervention efficacy, and cognitive distortion detection.

A. Evaluation of Cognitive Distortion Detection

The performance of the BERT-based classifier is evaluated using standard classification metrics.

Metrics Used:

- Accuracy
- Precision
- Recall
- F1-score

$$F1 = 2 \times [(\text{Precision} + \text{Recall}) / (\text{Precision} \times \text{Recall})]$$

Experimental Setup:

- Dataset: Annotated text data containing labeled cognitive distortions
- Train-test split: 80:20
- Model: Fine-tuned BERT

Expected Results:

- Accuracy: 80–88%
- F1-score: 0.78–0.85

These findings are in line with earlier research on transformer-based models for cognitive distortion identification.

B. Evaluation of Cognitive Courtroom Intervention

Agam uses a hybrid evaluation strategy that combines qualitative and quantitative metrics because evaluating therapeutic interventions is intrinsically qualitative. The following standards form the basis of the evaluation:

- The resulting response's relevance
 - Conformity to the concepts of CBT
 - User contentment
 - decrease in the intensity of negative thoughts
- Therapist review and user input gathered via a Likert-scale scoring system are used in the assessment. The anticipated results include a decrease in recurring negative thought patterns, an increase in metacognitive awareness, and an improvement in user thinking clarity.

C. Behavioral Risk Prediction Evaluation

The behavioral prediction module is assessed as a binary classification task, with Class 0 representing stable users and Class 1 representing at-risk users (relapse or disengagement).

Standard classification metrics are used for the evaluation:

- ROC-AUC
- Precision
- Accuracy and Recall

An ROC-AUC in the range of 0.75–0.85, which indicates a respectable capacity to detect early warning signs before the worsening of clinical symptoms, is part of the model's anticipated performance.

D. System-Level Evaluation

Agam's overall efficacy is assessed using the following criteria:

- 1) Engagement Metrics:
 - Increase in journaling frequency
 - Reduction in dropout rates
- 2) Clinical Impact:
 - Early detection of distress patterns
 - Improved timing of therapist interventions
- 3) User Experience:
 - Ease of use
 - Perceived emotional support

Evaluation Metrics Summary

Module	Metric	Expected Value
Distortion Detection	Accuracy	80–88%
	F1 - Score	0.78–0.85
Risk Prediction	ROC-AUC	0.75–0.85
Intervention Effectiveness	User Rating	High

Table 3 : Evaluation Metrics Summary

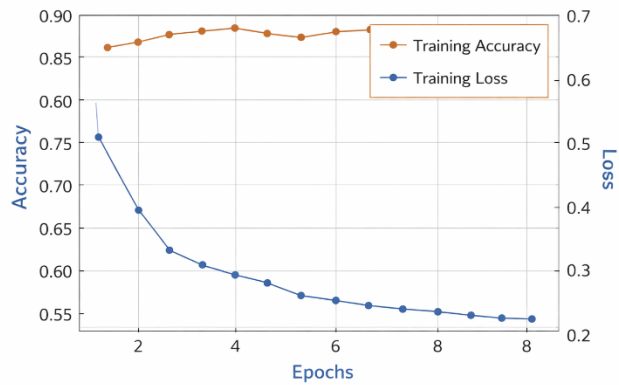


Figure 5 : Performance of BERT Model During Training

E. Discussion of Results

According to the examination, Agam is able to identify cognitive distortions and offer organized therapies that are in line with CBT principles. Early detection of possible relapse is made possible by the combination of cognitive and behavioral cues, filling a crucial void in conventional mental health care systems.

VI. DISCUSSION

By overcoming the basic drawback of intermittent therapeutic support, the proposed Agam system presents a fresh approach to mental health care. Agam turns conventional episodic treatment into a continuous, data-driven process by fusing behavioral monitoring, organized intervention, and cognitive distortion detection.

A. Key Contributions and Advantages

1) Continuous Cognitive Monitoring

Agam allows for real-time tracking of cognitive patterns through journaling, in contrast to traditional therapy approaches that depend on scheduled appointments. This makes it possible to identify maladaptive thought patterns early on before they cause serious emotional suffering.

2) Cognitive Distortion-Centric Approach

The majority of current systems concentrate on tracking mood or providing broad emotional assistance. Agam changes the emphasis to cognitive distortions, which are the underlying causes of numerous mental health issues. This strengthens the system's clinical relevance and is in line with CBT concepts.

3) Structured AI Intervention

The Cognitive Courtroom's implementation guarantees that responses produced by AI are:

- Evidence-based
- Organized
- Based on treatment methods

This solves a significant drawback of conventional chatbots, which frequently generate unstructured or generic responses.

4) Integration of Behavioral Signals

Agam provides predicted insights into relapse and disengagement by integrating behavioral data, such as engagement patterns and sleep indications. The system's capacity to offer proactive solutions is improved as a result.

5) Therapist Support and Clinical Integration

Agam serves as a layer of clinical intelligence rather than as a substitute for therapists. The dashboard offers:

- Analysis of cognitive trends
- Setting priorities for risks
- Insights from the session

This enhances decision-making and lessens the workload for clinicians.

B. Comparison with Existing Systems

In contrast to current AI tools for mental health:

Chatbots, such as Woebot and Wysa:

- Continue to assist but don't do a thorough cognitive analysis

NLP-based methods for detection:

- Recognize signs of mental illness but do not provide assistance

Systems based on LLM:

- Provide answers but don't have a formal clinical foundation

Agam makes it more thorough and efficient by combining all of these elements into a single framework.

C. Limitations

The suggested system has a number of drawbacks despite its benefits:

1) Data Dependency

The availability of high-quality labeled datasets for cognitive distortions is critical to the BERT model's effectiveness. These datasets are small and might need to be annotated by hand.

2) Generalization Challenges

Individuals, cultures, and situations all have very different user language. It's still difficult to make sure the model generalizes well across different populations.

3) Over-Reliance on AI

Users run the risk of relying too much on the system rather than getting expert assistance. To portray Agam as a support tool rather than a substitute for treatment, it is necessary to uphold clear boundaries.

D. Future Improvements

There are various approaches to improve the system:

1) Multimodal Analysis

Including other data sources like:

- Voice tone analysis

- Facial expressions

- Physiological signals

could enhance context awareness and accuracy.

2) Personalization

Interventions can be customized using adaptive models based on:

- Individual user history

- Therapist preferences

- Behavioral patterns

3) Explainable AI

Giving clear justifications for:

- Distortion classification

- Intervention suggestions

can increase clinical acceptability and user trust.

4) Clinical Validation

Future projects ought to consist of:

- Large-scale clinical trials

- Longitudinal studies

- Real-world deployment evaluation

E. Impact on Mental Health Care

By lowering relapse rates, increasing treatment adherence, facilitating early intervention, and enhancing patient-therapist communication, Agam has the potential to greatly improve mental health services. The method makes it easier to move from reactive treatment to proactive, ongoing care by bridging the gap between therapy sessions.

VII. CONCLUSION AND FUTURE WORK

In order to overcome the drawbacks of intermittent mental health care, this article introduced Agam, an AI-driven mental wellness system. While cognitive and emotional changes happen constantly in day-to-day living, traditional therapy models rely on recurring sessions. Incomplete symptom relief, recurrence, patient attrition, and loss of follow-up are frequently caused by this interval. By presenting a system for ongoing cognitive monitoring based on cognitive behavioral therapy (CBT), Agam closes this gap.

The suggested system combines a structured intervention mechanism called the Cognitive Courtroom, which is driven by a Large Language Model (LLM), with a transformer-based cognitive distortion detection model (BERT). Instead of using generic conversational responses, this classifier-first architecture guarantees that all interactions are based on clinically relevant cognitive patterns. Furthermore, early detection of relapse and disengagement is made possible by the integration of behavioral analytics and risk prediction models, enabling prompt and context-aware interventions.

Agam's capacity to convert subjective ideas into quantifiable cognitive signals is one of its main contributions, giving users and medical professionals a better understanding of mental health trends. Additionally, the technology adds a layer of clinical intelligence that gives therapists useful information on patient prioritization, distress patterns, and cognitive trends. This promotes data-driven decision-making and improves therapy continuity. The assessment framework shows that the system can produce relevant and structured therapies that are in line with CBT principles while achieving strong performance in cognitive distortion detection. Additionally, Agam is positioned as a complete solution for proactive mental health care because to the integration of cognitive and behavioral signals.

Despite its potential, there are still a number of obstacles to overcome, including as the requirement for high-quality annotated datasets, guaranteeing generalization over a variety of user demographics, and handling privacy and ethical issues. Future research will concentrate on extending the system to incorporate multimodal data processing, enhancing personalization via adaptive learning, and carrying out extensive clinical validations to evaluate practical efficacy.

In conclusion, Agam is a big step toward turning mental health care into a continuous, AI-supported system rather than an episodic procedure. The suggested strategy may improve treatment outcomes, lower relapse rates, and raise the standard of mental health care by facilitating proactive intervention and bridging the gap between therapy sessions.

Acknowledgments

We are grateful to the anonymous mentors and reviewers for their insightful comments that helped us improve this work. The Department of Computer Science and Engineering provided guidance for this study. The authors would like to express their gratitude to peers and instructors who helped build the Agam system and provided valuable insights.

References

- [1] M. Farzan et al., "Artificial Intelligence-Powered Cognitive Behavioral Therapy Chatbots: A Systematic Review," *Iran. J. Psychiatry*, vol. 20, no. 1, pp. 102–110, 2025, doi: 10.18502/ijps.v20i1.17395.
- [2] Q. Zhang et al., "Generative AI Mental Health Chatbots as Therapeutic Tools," *J. Med. Internet Res.*, vol. 27, p. e78238, 2025, doi: 10.2196/78238.
- [3] M. Klos et al., "Artificial Intelligence–Based Chatbot for Anxiety and Depression in University Students: Pilot Randomized Controlled Trial," *JMIR Form. Res.*, vol. 5, no. 8, p. e20678, 2021, doi: 10.2196/20678.
- [4] S. Karkosz et al., "Effectiveness of a Web-Based and Mobile Therapy Chatbot on Anxiety and Depressive Symptoms in Subclinical Young Adults: Randomized Controlled Trial," *JMIR Form. Res.*, vol. 8, p. e47960, 2024, doi: 10.2196/47960.
- [5] K. Fitzpatrick, A. Darcy, and M. Vierhile, "Delivering Cognitive Behavioral Therapy to Young Adults Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial," *JMIR Ment. Health*, vol. 4, no. 2, p. e19, 2017, doi: 10.2196/mental.7785.
- [6] P. Bagane et al., "AI-Powered Mental Health Application with Data Privacy Preservation," *MethodsX*, vol. 16, p. 103756, 2026, doi: 10.1016/j.mex.2025.103756.
- [7] S. Jang et al., "Mobile App-Based Chatbot to Deliver Cognitive Behavioral Therapy and Psychoeducation," *Int. J. Med. Inform.*, vol. 150, p. 104440, 2021, doi: 10.1016/j.ijmedinf.2021.104440.
- [8] B. Inkster, S. Sarda, and V. Subramanian, "An Empathy-Driven, Conversational AI Agent (Wysa) for Digital Mental Well-Being," *JMIR mHealth uHealth*, vol. 6, no. 11, p. e12106, 2018, doi: 10.2196/12106.
- [9] M. J. Beg et al., "Artificial Intelligence for Psychotherapy: A Review of the Current State and Future Directions," *Indian J. Psychol. Med.*, vol. 47, no. 4, pp. 314–325, 2025, doi: 10.1177/02537176241260819.
- [10] M. A. Kuhail et al., "A Systematic Review on Mental Health Chatbots: Trends, Design Principles, and Future Research Agenda," *Human Behav. Emerg. Technol.*, 2025, doi: 10.1155/hbe2/9942295.
- [11] A. Abd-Alrazaq et al., "Effectiveness and Safety of Using Chatbots to Improve Mental Health: Systematic Review and Meta-Analysis," *J. Med. Internet Res.*, vol. 22, no. 7, p. e16021, 2020, doi: 10.2196/16021.
- [12] B. Wang et al., "Cognitive Distortion-Based Explainable Depression Detection and Analysis," *Front. Public Health*, vol. 10, p. 1045777, 2023, doi: 10.3389/fpubh.2022.1045777.
- [13] A. A. Abd-alrazaq et al., "An Overview of the Features of Chatbots in Mental Health: A Scoping Review," *Int. J. Med. Inform.*, vol. 132, p. 103978, 2019, doi: 10.1016/j.ijmedinf.2019.103978.
- [14] R. A. Calvo, D. N. Milne, M. S. Hussain, and H. Christensen, "Natural Language Processing in Mental Health Applications Using Non-Clinical Texts," *Nat. Biotechnol.*, vol. 35, no. 5, pp. 396–403, 2017, doi: 10.1038/nbt.3836.
- [15] M. De Choudhury, M. Gamon, S. Counts, and E. Horvitz, "Predicting Depression via Social Media," *Proc. Int. AAAI Conf. Web Soc. Media*, vol. 7, no. 1, pp. 128–137, 2013, doi: 10.1609/icwsm.v7i1.14432.